

OpenClaw AI Script Writing for Peripheral Control Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-OpenClaw Prompts], [05-AI Voice Interaction Configuration], and [06-Three Conversation Scheme Configuration and Testing] in the `Preparation for use` directory. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only expands on the "Let AI generate independently runnable scripts" part.

1. Tutorial Objectives

This tutorial demonstrates through the method of "natural language generating hardware scripts" that OpenClaw is not only capable of directly controlling peripherals but can also help you save control logic into script files. After completing this tutorial, readers will be able to:

- Understand the basic idea of OpenClaw generating hardware control scripts
- Interact with OpenClaw via three methods: WhatsApp, TUI, and voice code
- Generate four common types of scripts: RGB, Servo, Temperature/Humidity, OLED

2. Hardware Preparation

- OpenClaw main board
- Servo / RGB light strip / OLED / DHT temperature and humidity sensor
- `[Optional] AI voice module + speaker`

Description:

- This tutorial does not require all peripherals to be connected at once
- Connect the corresponding hardware first for the type of script you plan to demonstrate

3. Software Preparation

Code entry points used in the tutorial

- Script generation skill: `/home/sunrise/.openclaw/workspace/skills/script-runner/SKILL.md`
- Default script save directory: `/home/sunrise/Desktop/`
- Hardware command example directory: `/home/sunrise/hardware_ha1_demo/`
- Voice entry point: `/home/sunrise/voice_to_openclaw/src/main.py`

The core of this capability is not "executing an instant command," but letting OpenClaw help you generate an independently runnable Python script. According to existing skill conventions:

- When users say "generate a script," "write a script," or "save to desktop," it will prioritize using `script-runner`
- Hardware scripts default to using `openclaw_lib`
- Default serial port is `/dev/ttyS1`
- Default baud rate is `115200`
- Scripts are usually saved in `/home/sunrise/Desktop/`

You can understand it as two usage methods:

- Method 1: Generate a script and run it immediately
- Method 2: Only generate the script, save it to the desktop first, and modify or re-run it later

4. Testing Script Generation Capability

Before starting this tutorial, please complete a basic `openclaw tui` conversation self-check first; if you haven't done so, see [05-Three Conversation Scheme Configuration and Testing]. After confirming that OpenClaw can respond normally, proceed to the script generation specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```

[[[OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_user_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

```

It is recommended to start with the simplest script requirements, for example:

- Help me write a script to make the RGB red light flash 5 times, save to desktop and run
- Generate a script to read temperature and humidity, do not run it yet
- Write a script to make servo 1 swing back and forth between 0, 90, and 180 degrees
- Generate a script to display "Hello OpenClaw" on the OLED

5. Three Schemes for Conversing with OpenClaw

Scheme	Suitable Scenarios	Advantages	Suggested Positioning
WhatsApp	Remote requirement submission, classroom demonstration	Can directly describe "help me write what script"	Used to demonstrate natural language generation capability
TUI	Local debugging, verifying generation results	Easiest to see script generation and execution return	Preferred debugging scheme
Voice Code	Voice interaction, complete AI experience	Can directly orally describe script requirements	As a complete demonstration scheme

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to enable broadcast parameter

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal.
```

- Enter in terminal:

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Scheme

For WhatsApp access steps, please refer directly to [06-Three Conversation Scheme Configuration and Testing]. This section only retains conversation examples suitable for the script generation tutorial.

You can converse directly, for example:

- Generate a script for all servos to swing left and right for 5 seconds and run it

20:07



< 2



+86 153 3885 7526 (你)

给自己发消息

HTTP 400: Access denied, please make sure your account is in good standing. For details, see: <https://help.aliyun.com/zh/model-studio/error-code#overdue-payment>

19:59 ✓✓

Generate a script that makes servo swing left and right for 10 seconds and run it

20:07 ✓✓

Done! Generated `servo_swing.py` on the desktop and executed it successfully.

The servo swung left (0°) and right (180°) repeatedly for 10 seconds, then returned to the center position (90°).

20:07 ✓✓



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123



space

return

5.2 TUI Scheme

If you have completed the TUI self-check in Part 6, you can continue the conversation by entering the following command in the terminal:

```
openclaw tui
```

Then you can directly input script requirements, for example:

- Help me write a script to control RGB to become a blue breathing light and run for 10 seconds

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

帮我写个脚本控制 RGB 变成蓝色呼吸灯运行10秒

已生成脚本 blue_breathe.py, 保存在桌面, 运行成功: 蓝色呼 吸灯运行 10 秒后自动关闭。✳
connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 50k/1.0m (5%)
```

5.3 Voice Code Scheme

Requires selection of AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [06-AI Voice Interaction Configuration]. After wake-up, you can directly speak control content, for example:

- Write a script to display current temperature and humidity on the screen and run it



The screenshot shows a desktop environment with a file explorer on the left and a terminal window on the right. The file explorer displays two files: 'blue_breathe.py' and 'temp_humidity_display.py'. The terminal window shows the following text:

```
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Write a script that displays current temperature and humidity on the screen and run it.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! I created and ran the script `temp_humidity_display.py` on your desktop. 🎉

The script:
1. Read the current temperature:
   **30.3°C**
2. Read the current humidity: **50.0%**
3. Displayed both
   values on the OLED screen for 15 seconds
4. Cleared the screen
   automatically

The display is now cleared. The script is saved on
your desktop if you want to run it again later!
[信息] 回复已裁剪为播报文本: 372 -> 76 字
```

6. Comprehensive Cases

Note: The following are conversation demonstrations conducted via `openclaw_tui`. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: RGB Flashing Script

Experiment Objective

Let OpenClaw generate a basic lighting script to verify the "script creation + peripheral execution" pipeline.

Effect Description

OpenClaw will organize your natural language requirements into a Python script. After the script is saved to the desktop, it can be run directly or kept for subsequent modification.

Example Instructions

- Help me write a script to make the RGB red light flash 5 times
- Generate a blue breathing light script, save to desktop and run
- Write a script to make the RGB perform a rainbow light effect

6.2 Case 2: Servo Action Script

Experiment Objective

Generate a servo action script to solidify single-dialogue actions into a repeatedly runnable program.

Effect Description

This type of script is suitable for actions like "nodding," "shaking head," "swinging back and forth," or "returning to center," and is also suitable for subsequent expansion into classroom demonstrations or combination projects.

Example Instructions

- Write a script to make servo 1 move back and forth between 0, 90, and 180 degrees
- Generate a servo nodding script
- Help me write a script to make all four servos return to 90 degrees together

6.3 Case 3: Temperature and Humidity Reading Script

Experiment Objective

Generate a script to read current temperature and humidity and output the results.

Effect Description

This type of script is suitable for sensor self-checks, data sampling, or subsequent log recording, and is also one of the most common types of basic scripts for making scheduled tasks later.

Example Instructions

- Generate a script to read temperature and humidity and save it to the desktop
- Help me write a script to print current temperature and humidity
- Write a script to read DHT data once and run it

6.4 Case 4: OLED Display Script

Experiment Objective

Generate a script to display text or metric content on the OLED.

Effect Description

This case is suitable for making welcome pages, status pages, or simple dashboards, and is also the basis for subsequent linkage with capabilities such as temperature/humidity and cameras.

Example Instructions

- Help me write a script to display "Hello OpenClaw" on the OLED
- Generate a script to display temperature and humidity on the OLED
- Write a script to display a page of simple status information

6.5 Case 5: Combination Script

Experiment Objective

Let OpenClaw generate a comprehensive script containing multiple peripheral actions at once.

Effect Description

This type of script can string together "reading sensors," "displaying on OLED," "controlling RGB," and "driving servos," suitable for transitioning from single-peripheral control to small project scripts.

Example Instructions

- Write a comprehensive script: read temperature and humidity then display on OLED
- Generate a script: first light up green light, then make the servo turn to 90 degrees
- Help me write a script: light up red light and prompt on OLED if temperature is too high